

OI-26-731 / Feeder 12R incident record

MetroGrid Electric | Portland North district | Event 2 July 2026

CLOSED

EVENT RECORD

Field	Recorded value
Outage start	2 Jul 2026 15:42
Customers interrupted	8,960
Critical customers	Hospital loop; traffic signal TS-12; lift station LS-4
Hospital restoration	17:18 via tie T-7
All-customer restoration	19:06 via R12 breaker close
Network normalized	19:32 after T-7 opened

INCIDENT SOURCE REGISTER

Record	Clock basis	Custodian	State
SCADA alarm export	Sub-second	System control	Frozen 20:00
Switching order 12R-731	24-hour local	Control room	Signed
OMS restoration series	5-minute bins	Outage management	Final
Patrol media P1-P3	GPS device time	Crew A	Uploaded
Investigation report	Revision 2	Protection engineering	Signed 3 Jul

SCADA alarm sequence

Sequence exported from control historian at 20:00 local

Time	Device	State	System note
15:42:11	R12 breaker	Trip	Phase B ground
15:42:14	Recloser 12R-3	Lockout	Third shot
15:42:19	Relay 50G	Pickup	7.8 kA momentary
15:43:02	Cap bank CB-4	Offline	Voltage sag
15:44:33	OMS	8,960 out	Nested outage created
16:05:20	Switch S-18	Opened	Crew isolation
16:31:05	Switch S-22	Opened	Riverside isolated
17:11:00	DER-44	0 kW	Curtailement verified
17:18:44	Tie T-7	Closed	Hospital backfeed
18:05:02	F-12B	Replaced	Maple test passed
18:55:18	Switch S-18	Closed	Upstream path staged
19:00:41	Switch S-22	Closed	Riverside path staged
19:06:10	R12 breaker	Closed	Normal source restored

Timestamps are historian event times; switching-order times on the following page are minute-level operator entries.

Switching order 12R-731

Operator entries and cumulative customers restored

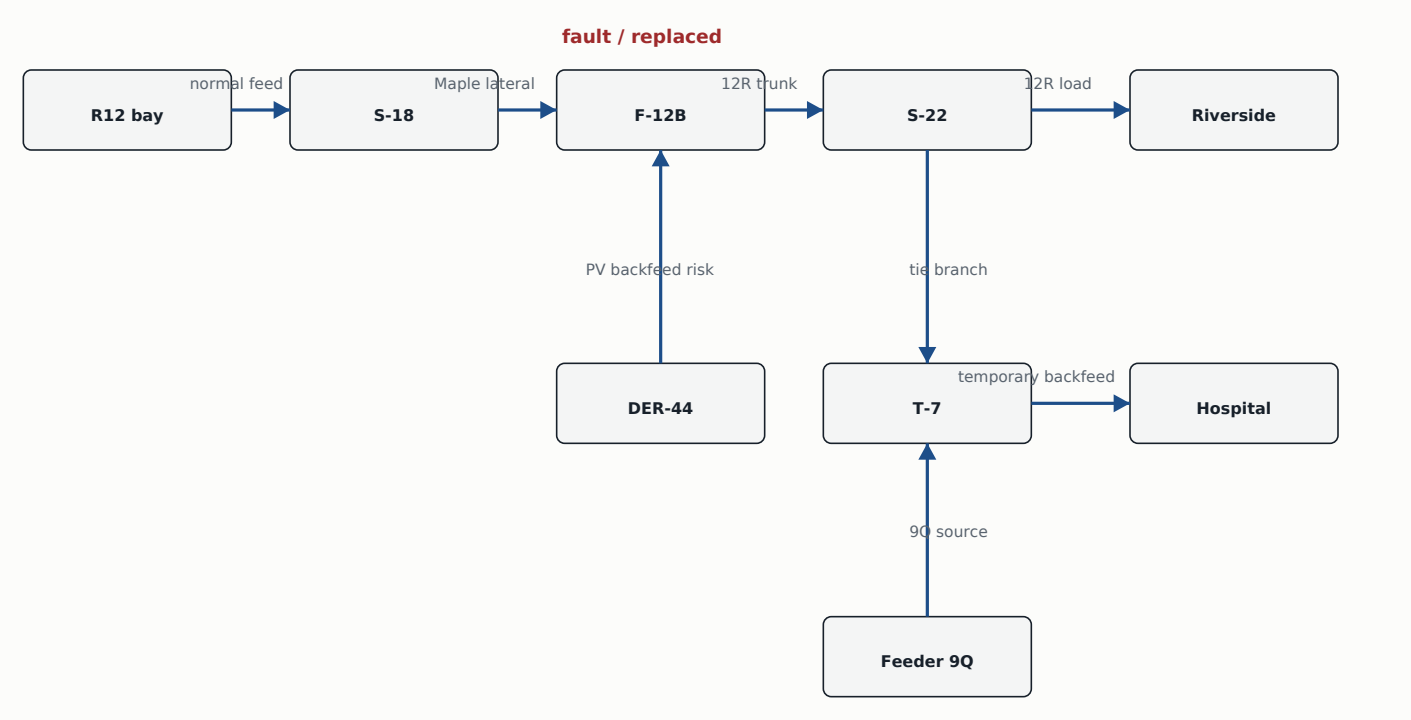
Step	Time	Action	Authority	Cumulative restored
1	16:05	Open S-18	Crew A / Control	0
2	16:31	Open S-22	Crew B / Control	0
3	16:47	Test Maple lateral	Crew A	0
4	17:11	Verify DER-44 at 0 kW	System control	0
5	17:18	Close tie T-7	Crew C / Control	4,820
6	18:05	Replace and test F-12B	Crew A	6,140
7	18:42	Patrol Riverside taps	Crew B	6,140
8	18:55	Close S-18	Crew A / Control	6,140
9	19:00	Close S-22	Crew B / Control	6,140
10	19:06	Close R12 breaker	System control	8,960
11	19:32	Open T-7 / normalize	Crew C / Control	8,960 stable

ORDER SIGNOFF

Role	Name	Signed
Switching authority	N. Ortega	2 Jul 19:40
Field lead	R. Mehta	2 Jul 19:44
Control desk	A. Patel	2 Jul 19:45

Feeder one-line and DER clearance

Normal feeder path, outage isolation, and temporary hospital backfeed

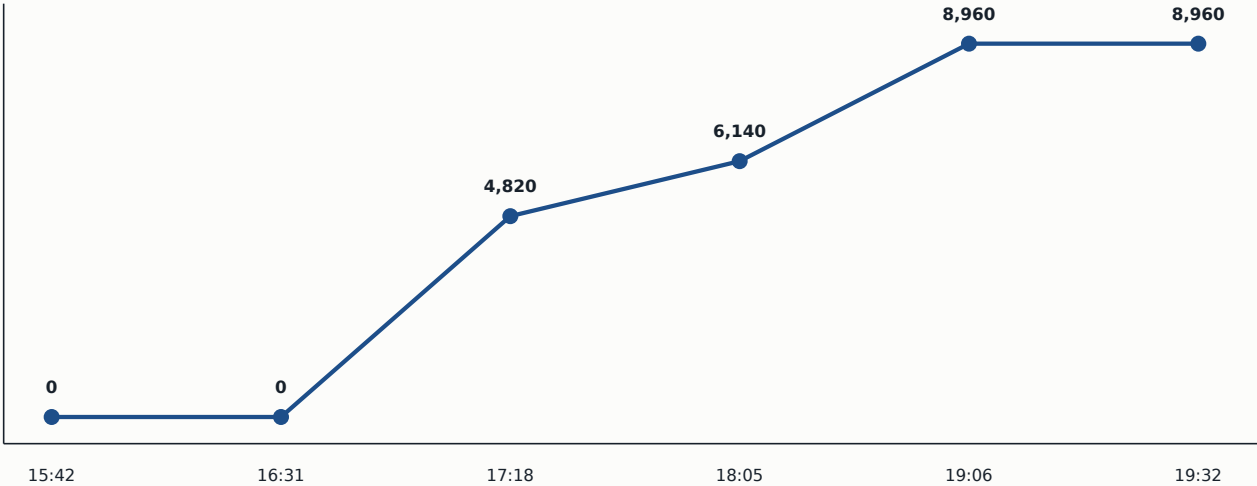


DER AND TRANSFER CLEARANCE

Clearance item	Operator entry	Field confirmation	State
DER-44 disconnect	Open 17:06	Visible at POI	Accepted
SCADA output	0 kW 17:11	Stable 5 min	Accepted
Feeder 9Q load	74% after transfer	Below emergency rating	Accepted
Reverse power	None	Relay 67P clear	Accepted
Hospital ATS	Normal source lost	T-7 source accepted	Accepted
T-7 permission	Issued 17:16	Close at 17:18	Accepted
DER restore	After normalize	Site call 19:34	Deferred

Restoration progression and customer impact

OMS final series; customer restoration ends at 19:06



CUSTOMER IMPACT

Segment	Customers	Critical load	Restored
9Q transfer block	4,820	Includes hospital + TS-12	17:18
Hospital loop	1,240	Hospital	17:18
TS-12	1 service	Traffic signal	17:18
Maple tested block	1,320	Includes care home spur	18:05
Care home spur	86	Assisted living	18:05
Riverside / industrial block	2,820	Includes LS-4	19:06
LS-4	1 service	Lift station	19:06

Parent blocks reconcile to 8,960; critical-service rows are nested callouts and are not added again.

Patrol field media

Crew A uploads with GPS device time; original files retained in incident archive



P1 F-12B / 16:18

Blackened split polymer barrel is visible below the phase-B conductor. The red review circle is on the damaged cutout.



P2 Maple span / 16:23

The span between the two poles is clear. Vegetation is set back and no branch touches either conductor.



P3 S-18 / 16:31

The switch blade is visibly open and the yellow S-18 OPEN hold tag is attached below the blade.

Draft investigation report / reviewer markup

Revision 1 retained in the incident file; signed Revision 2 follows

OUTAGE REPORT - WORKING COPY

METROGRID OUTAGE INVESTIGATION / REVISION 1

Event OI-26-731 / Feeder 12R / prepared 02 Jul 21:10

Preliminary cause: TREE CONTACT on Maple lateral.

Hospital loop restored: 18:05.

DER-44 status: not recorded.

Weather: 46 mph gust; primary cause pending review.

F-12B

17:18

PROTECTION REVIEW / 03 JUL 08:35

Cause revised to failed polymer cutout F-12B.

Hospital restored 17:18 via T-7; 18:05 is Maple lateral.

DER-44 curtailed to 0 kW at 17:11 before tie close.

Weather retained as context; P2 rules out tree contact.

REVIEWER EVIDENCE MATRIX

Evidence	Observed state	Use in Revision 2
Photo P1	Split / char at F-12B	Supports equipment failure
Photo P2	Clear span / no contact	Rejects tree-contact draft
Relay 50G	Single phase-ground impulse	Matches failed cutout
Cutout tag F12B-882	Polymer barrel split	Asset-lab teardown
Lightning log	None within 5 miles	Not causal

Revision 2 final investigation and notice

Signed 3 July 2026 | controlling incident determination

EQUIPMENT FAILURE

FINAL DETERMINATION

Control field	Final value	Basis
Cause	Failed polymer cutout F-12B	P1 + relay + tag F12B-882
Hospital restoration	17:18	T-7 switching step 5
All customers restored	19:06	R12 close / OMS
DER-44	Curtailed 17:11	0 kW before T-7 close
Major-event threshold	No	8,960 customers / 3 h 24 m
Follow-up	WO-7714	Replace peer cutouts / lab teardown

NOTICE ROUTING

Recipient	Channel	Due / sent	Payload
Hospital liaison	Phone	Sent 17:21	Backfeed restored
City OEM	Email	Sent 18:12	Critical-load update
State PUC log	Portal	Due 05 Jul 12:00	Below major threshold
DER owner	Phone	Sent 19:34	Normalization complete
Asset engineering	Work order	Sent 03 Jul 09:05	F12B-882 teardown
Vegetation contractor	No dispatch	Not required	P2 rules out tree contact

FINAL SIGNOFF

- ☒ Operations sequence / N. Ortega 09:12
- ☒ Protection / R. Mehta 10:44
- ☒ Regulatory / S. Kim 11:20
- ☐ Vegetation / not required